

### 11.20.3.1. Determining the OTA Provider to query

The discovery of available OTA Providers is necessary for OTA Requestors to be able to query for new software images. Each OTA Requestor SHALL keep a list of OTA Provider Operational Identifiers (see [Section 2.5.5, “Node Identifier”](#)) that it could query.

A given OTA Requestor SHALL have sufficient storage to maintain one OTA Provider entry per Fabric within the [DefaultOTAProviders](#) default list. This default OTA Provider list MAY be augmented by any means deemed acceptable by a given OTA Requestor, such that the internal list of possible locations to query contains at least the DefaultOTAProviders, but it MAY contain more. For example, it may contain cached locations that arose from the [AnnounceOTAProvider](#) command.

When an OTA Requestor determines that it is time to query an OTA Provider, it MAY use any method of its choosing to determine which OTA Provider to contact for its next query.

An OTA Requestor MAY expunge OTA Providers from its OTA Provider list if it determines that the entry is stale or obsolete.

Discovery of additional OTA Providers MAY be done by handling the arrival of [AnnounceOTAProvider](#) commands invoked by OTA Providers.

Commissioners SHOULD add an entry to the [DefaultOTAProviders](#) list attribute, if an OTA Provider is known at commissioning time, to reduce the delay between commissioning and first [QueryImage](#) command.

Whenever communicating with an OTA Provider location obtained either through the DefaultOTAProviders attribute, or the AnnounceOTAProvider command, an OTA Requestor SHALL target all interactions with that Node by interacting with the given Endpoint on the given ProviderNodeID obtained from these sources.

Discovery of additional OTA Providers MAY be done using runtime service discovery, which is outside the scope of this specification.

Nodes MAY attempt to contact OTA Providers that are known to them in any order if failing to reach a default OTA Provider from an entry in the [DefaultOTAProviders](#) list. This approach would assist in maximizing likelihood of eventual success.

### 11.20.3.2. Querying the OTA Provider

Query of the OTA Provider SHALL be done using the [QueryImage](#) command. The arguments for this command provide sufficient information to allow the OTA Provider to determine the availability of a new image for the querying OTA Requestor.

An OTA Requestor SHALL NOT query more frequently than once every 120 seconds, unless a Node loses its timekeeping state, due to events such as power loss or restart, that prevent applying such a delay. This reduces the burden on both the OTA Providers providing the service to a large number of nodes and the supporting networking infrastructure. It is recommended for OTA Requestors to attempt a daily [QueryImage](#) command, if capable, to ensure timely access to updated software, including security-critical updates.

The OTA Provider SHALL use an algorithm deemed satisfactory by its implementer to determine